SUPPLEMENTARY MATERIAL

A. Pseudo Algorithms

Algorithm 1 provides the pseudo code of the meta-knowledge acquisition process:

Algorithm 1 Meta-Knowledge Acquisition

Require: Task distribution $p(\mathcal{T})$, simplified environment E_{simp}, encoder q_ϕ, decoder p_ξ, high-level policy π_{high}, DPMM prior $p_\theta(z)$

```
Initialize q_φ, p_ξ, p_θ(z), π_high
for each training iteration do
    Sample task T_i ~ p(T)
    Reset E_simp with T_i
    Rollout {τ = {(s_t, a_t, r_t, s_{t+1})}_{t=0}^T}_{k=1}^K ~ π_high
    Construct context windows c_t with length H
    for each context window c_t do
        z_t ← q_φ(· | c_t)
        (ŝ_{t+1}, r̂_t, ŷ_t) ← p_ξ(z_t, a_t^high, s_t)
        Compute L_rec, D_KL (Eq. 19, 20)
    end for
    Update I(φ, ξ) ← ∇L_I
    z̄_t = sg(z_t), compute L_π^high (Eq. 22)
    Update π_high ← ∇L_π^high
    if time for DPMM update then
        Update p_θ(z) ← Z = {z_n}_{n=1}^{N_z}, birth or merge components according ELBO (Eq. 14)
    end if
end for
return trained I_θ = (q_φ, p_ψ), π_high
```

Algorithm 2 presents the overall learning and inference procedure of the proposed framework.

B. Necessity of Stable Low-level Control

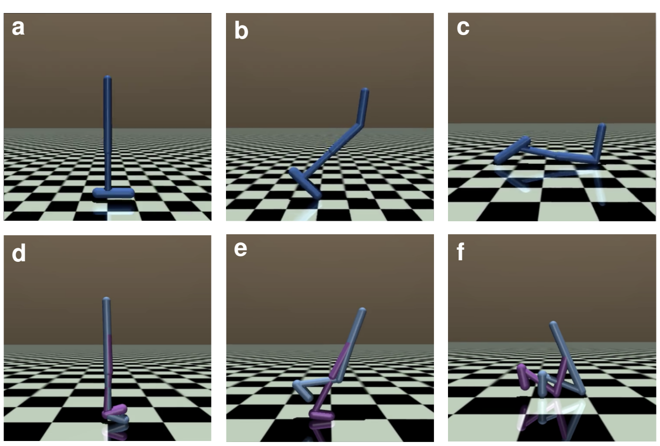

Fig. S1: Early-stage instability of complex locomotion agents under insufficient low-level exploration. **a–c**, Hopper initially maintains an upright posture, but its center of gravity gradually moves outside the support region, leading to falling. **d–f**, Walker exhibits a similar failure process, where unstable body posture and delayed recovery cause the agent to collapse. These examples motivate the need for a disentangled low-level warm-up (Q2) before reusing task-level meta-knowledge.

Algorithm 2 ReMAP

Require: $p(\mathcal{T})$, E_{simp}, target agent set $\{\mathcal{A}_i\}_{i=1}^N$

Ensure: Reusable meta-knowledge module $\mathcal{M}_{\text{meta}}$ and low-level policies $\{\pi_{\text{low}}^{(i)}\}_{i=1}^N$

```
// Stage 1: Meta-Knowledge Acquisition
Train I = (q_φ, p_ξ) and π_high on E_simp using Algorithm 1
Construct M_meta ← (I, π_high) according to Eq. 23
// Stage 2: SMAI-guided Disentangled Learning
for each target agent A_i do
    Train π_low^(i)(·|s_t^(i), g_t) with SMAI in Eq. 25
    Apply curriculum schedules in Eqs. 28–32
end for
// Stage 3: Cross-Agent Meta-Knowledge Reutilization
Freeze I, π_high, and {π_low^(i)}_{i=1}^N
for each target agent A_i do
    Initialize stride predictor P_ζ^(i)
    for each interaction episode do
        for each high-level decision step t do
            Observe s_t^(i) and compute s̄_t^(i) using Eq. 33
            Construct c̄_t^(i) using Eq. 34
            Infer z_t^(i) using Eq. 35
            Generate a_t^{high,(i)} using Eq. 36
            Compute α_t^(i) and m_t^(i) using Eqs. 37–38
            Obtain b_t^(i) using Eq. 39
            Construct g_t^(i) using Eq. ??
            Predict k_t^(i) ← P_ζ^(i)(g_t^(i))
            for τ = t to t + k_t^(i) − 1 do
                Execute a_τ^(i) ~ π_low^(i)(·|s_τ^(i), g_t^(i))
            end for
            Update P_ζ^(i) using Eq. 40
            Append mapped transition to the context buffer
        end for
    end for
end for
return M_meta, {π_low^(i)}_{i=1}^N
```

The necessity of stable low-level control is further illustrated in Fig. S1. This figure shows a common failure mode of end-to-end training during cold-start exploration. For unstable agents such as Hopper and Walker, the center of gravity can quickly move outside the support region, causing the agent to collapse. Once the agent falls, it can no longer collect meaningful locomotion trajectories for either task inference or policy improvement. In this case, even if the task inference module could provide correct task semantics and the high-level policy could generate reasonable subgoal magnitudes, the overall system would still fail because the low-level policy is physically unable to realize the intended motion. Therefore, unstable low-level exploration directly prevents effective task-level learning and deployment.

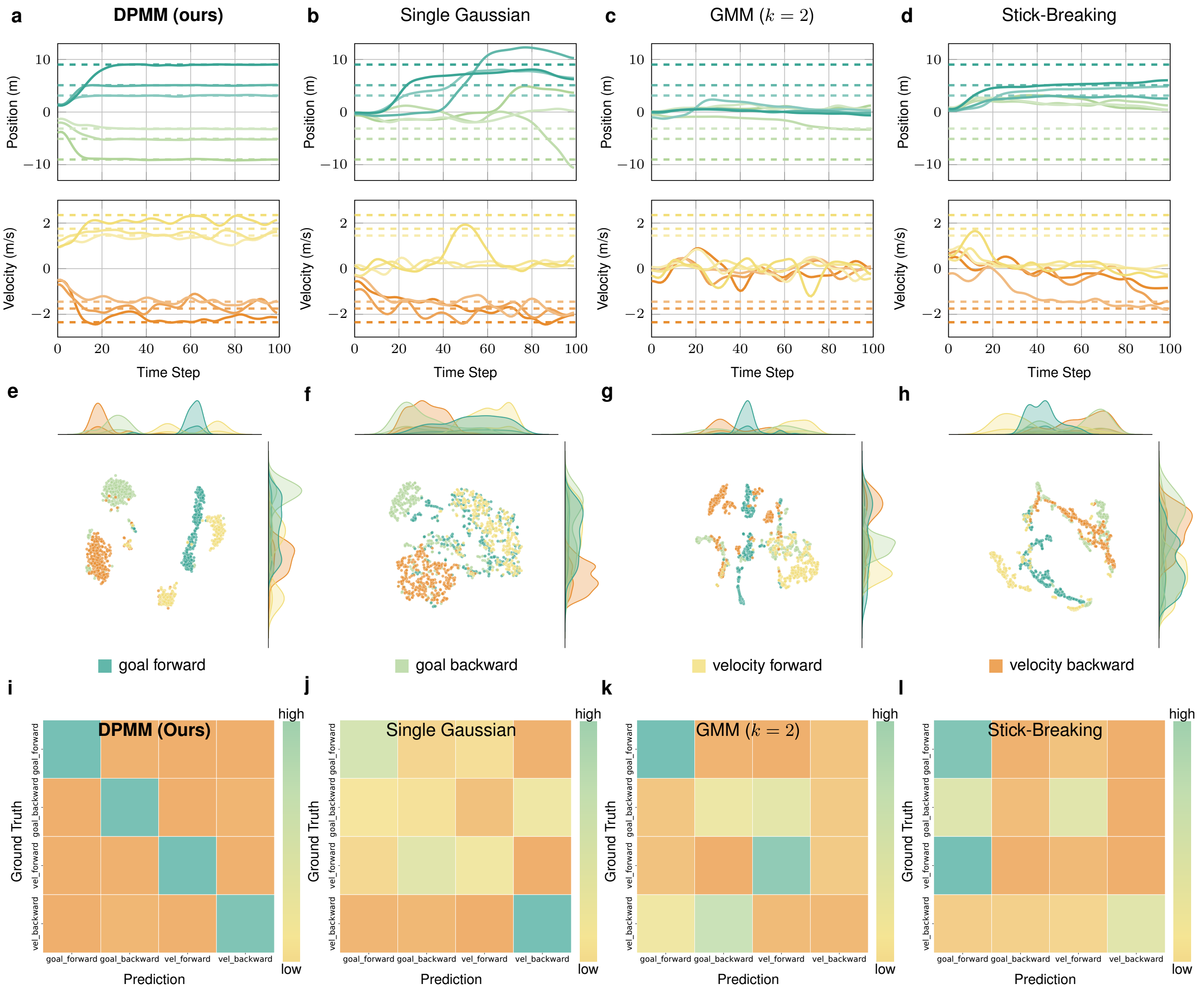

Fig. S2: Ablation comparison of different latent prior formulations on the Half-Cheetah non-parametric tasks. **a–d**, Inference trajectories of DPMM (ours), Single Gaussian, GMM with $k = 2$, and Stick-Breaking, respectively. For each panel, the upper plot shows position tracking and the lower plot shows velocity tracking across twelve non-parametric tasks; dashed horizontal lines denote the corresponding ground-truth targets. **e–h**, t-SNE visualizations of the learned latent representations under the four prior formulations. Our proposed DPMM prior produces more compact and better separated task-semantic clusters, while the ablated priors show stronger overlap between task categories. **i–l**, Decoder prediction confusion matrices under the four prior formulations. The more diagonal structure obtained by DPMM indicates that its inferred latent representations preserve clearer task semantics after deployment, whereas the increased confusion in the ablated variants shows that weaker latent organization directly degrades task inference.

C. Ablation Study

To analyze the contribution of the structured latent prior in the task inference module, we conduct an ablation study by replacing the DPMM prior with three alternative designs: a single Gaussian prior, a GMM prior with a fixed but improper number of components (in our case, $k = 2$), and a stick-breaking prior. All variants use the same disentangled training and deployment pipeline, and differ only in how the latent task space is regularized. This comparison examines whether accurate cross-embodiment reutilization requires an adaptive task-mode structure, rather than only a generic continuous latent representation.

Fig. S2 summarizes the ablation results on the Half-Cheetah embodiment. Figs. S2a–d show the tracking trajectories of the original model and the three ablated variants. The original model produces stable goal- and velocity-tracking behaviors across the four task types, indicating that the inferred task representation can provide correct semantic-magnitude guidance to the low-level controller. In contrast, the ablated variants exhibit larger tracking deviations or unstable responses. This suggests that when the latent prior cannot organize task modes properly, the high-level policy receives ambiguous task information and may generate inaccurate subgoals.

Figs. S2e–h further visualize the inferred latent representations using t-SNE. With the full model, different task types form compact and clearly separated clusters, showing that the DPMM prior helps organize the latent space according to task semantics. The single Gaussian prior tends to compress all task modes into one shared distribution, leading to mixed embeddings and weak task separation. The fixed-component GMM prior provides partial clustering, but its predefined component number imposes an improper structural assumption when it does not match the actual task modes. The stick-breaking prior introduces non-parametric flexibility, but the resulting embeddings are less compact and show less stable task boundaries in this setting.

The semantic reliability of the inferred representations is also reflected in the decoder prediction results in Figs. S2i–l. Our original model produces the clearest diagonal structure in the confusion matrix, indicating that the inferred latent variables remain strongly predictive of the correct task identities. The alternative priors introduce more off-diagonal confusion, especially between task categories with similar motion directions or magnitudes. These errors directly explain the degraded tracking performance in Figs. S2a–d, since incorrect task inference leads to inappropriate semantic-magnitude subgoals.

Overall, the ablation results show a consistent causal relation: the latent prior determines how task semantics are organized, this organization affects decoder-based task prediction, and the resulting task confidence and magnitude guidance determine downstream tracking performance. Compared with the single Gaussian, fixed-component GMM, and stick-breaking variants, the DPMM prior provides a more reliable task-mode structure for the learned meta-knowledge. This confirms that the proposed DPMM-regularized task inference module is a key component for learning compact, semantically separable, and reusable meta-knowledge under non-parametric task distributions.